Chapter 1 - Rings and Ideals

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Exercise 1

Let x be a nilpotent element of a ring A. Show that $1+x$ is a unit of A. Deduce that the sum of a nilpotent element and a unit is a unit.

Solution. Suppose $x^r = 0$. Notice

$$\begin{aligned}(1+x)\left(1-x+x^2-x^3+\ldots+(-1)^{r-1}x^{r-1}\right) &= 1-x^r \\ &= 1\end{aligned}$$

So $1+x$ is a unit in A.

Exercise 2

Let A be a ring and let $A[x]$ be the ring of polynomials in an indeterminate x, with coefficients in A. Let $f = a_0 + a_1x + \cdots + a_nx^n \in A[x]$. Prove that

(i) f is a unit in $A[x] \Leftrightarrow a_0$ is a unit in A and $a_1, \ldots, a_n$ are nilpotent. [If $b_0 + b_1x + \cdots + b_mx^m$ is the inverse of f, prove by induction on r that $a_n^{r+1}b_{m-r} = 0$. Hence show that a_n is nilpotent, and then use Ex. 1.]

(ii) f is nilpotent $\Leftrightarrow a_0, a_1, \ldots, a_n$ are nilpotent.

(iii) f is a zero-divisor $\Leftrightarrow$ there exists $a \neq 0$ in A such that $af = 0$. [Choose a polynomial $g = b_0 + b_1x + \cdots + b_mx^m$ of least degree m such that $fg = 0$. Then $a_nb_m = 0$, hence $a_ng = 0$ (because a_ng annihilates f and has degree $< m$). Now show by induction that $a_{n-r}g = 0$ $(0 \leq r \leq n)$.]

(iv) f is said to be primitive if $(a_0, a_1, \ldots, a_n) = (1)$. Prove that if $f, g \in A[x]$, then fg is primitive $\Leftrightarrow f$ and g are primitive.

Solution. (i) (a) ($\Rightarrow$): Let $g = b_0 + b_1x + \ldots + b_mx^m$ be the inverse of f. Now

$$(a_0 + a_1x + \ldots + a_nx^n)(b_0 + b_1x + \ldots + b_mx^m) = 1$$

By checking the coefficients of $x^{n+m}, x^{n+m-1}, x^{n+m-2} \ldots$, we get

$$\begin{cases} a_nb_m = 0 \\ a_n^2b_{m-1} = 0 \\ a_n^3b_{m-2} = 0 \\ \vdots \\ a_n^mb_1 = 0 \\ a_n^{m+1}b_0 = 0 \end{cases}$$

On the other hand, by checking the constant term directly,

$$a_0b_0 = 1$$

Therefore a_0 is unit, and $a_n^{m+1} = a_n^{m+1}b_0a_0 = 0$. So a_n is nilpotent.

Moreover, $-a_n x^n$ is a nilpotent element in $A[x]$. By *Ex.1*,

$$f - a_n x^n = a_0 + a_1 x + \ldots + a_{n-1} x^{n-1}$$

is a unit in $A[x]$, so by iterating the process above we'll see $a_{n-1}, \ldots, a_1$ are nilpotent.

(b) ($\Leftarrow$): By *Ex.1*, the sum of a unit and a nilpotent element is a unit. Notice $a_1 x, a_2 x^2, \ldots, a_n s^n$ are nilpotent in $A[x]$ while a_0 is a unit in $A[x]$, so their sum f is a unit in $A[x]$.

(ii) (a) ($\Rightarrow$) : Consider the constant term of power of f, from which we know a_0 is nilpotnet. Nilpotent elements are closed under addition, so

$$a_1 x + a_2 x^2 + \ldots + a_n x^n = f - a_0$$

is nilpotent, i.e.

$$a_1 + a_2 x + \ldots + a_n x^{n-1}$$

is nilpotent. By induction we can have $a_0, a_1, \ldots, a_n$ being nilpotent.

(b) ($\Leftarrow$) : When $a_0, a_1, \ldots, a_n$ are all nilpotent, let $k \in \mathbb{Z}^+$ be sufficiently large, and obviously we can finally reach $f^k = 0$.

(iii) (a) ($\Rightarrow$) : Suppose we have $g = b_0 + b_1 x + \ldots + b_m x^m$ such that $fg = 0$.
If $b_m = 0$, we're done. Otherwise, similar to *(i)*, by checking the coefficients of $x^{m+n}, x^{m+n-1}, \ldots$ respectively, we'll obtain

$$\begin{cases} a_n b_m = 0 \\ a_{n-1} b_m = 0 \\ a_{n-2} b_m^2 = 0 \\ \vdots \\ a_1 b_m^{n-1} = 0 \\ a_0 b_m^n = 0 \end{cases}$$

So $f \cdot b_m^n = 0$.

(b) ($\Leftarrow$) : Trivial.

(iv) (a) ($\Leftarrow$) : Let $g = b_0 + b_1 x + .. + b_m x^m$. Since g is primitive, $\exists \lambda_0, \lambda_1, \ldots, \lambda_m \in A, s.t.$

$$\lambda_0 b_0 + \lambda_1 b_1 + \ldots + \lambda_m b_m = 1$$

Therefore

$$\lambda_0 b_0 f + \lambda_1 b_1 f + \ldots + \lambda_m b_m f = f$$

Also, suppose

$$\mu_0 a_0 + \mu_1 a_1 + \ldots + \mu_n a_n = 1$$

Since

$$\begin{aligned} fg = \lambda_0\mu_0 b_0 a_0 &+ \lambda_1\mu_0 b_1 a_0 x + \lambda_2\mu_0 b_2 a_0 x^2 + \ldots + \lambda_m\mu_n b_m a_n x^{m+n} \\ &+ \lambda_0\mu_1 b_0 a_1 x + \lambda_1\mu_1 b_1 a_1 x^2 + \ldots \\ &+ \lambda_0\mu_2 b_0 a_2 x^2 + \ldots \\ &\vdots \end{aligned}$$

So

$$(\lambda_0\mu_0 b_0 a_0) + (\lambda_0\mu_1 b_0 a_1 + \lambda_1\mu_0 b_1 a_0) + (\lambda_0\mu_2 b_0 a_2 + \lambda_1\mu_1 b_1 a_1 + \lambda_2\mu_0 b_2 a_0) + \ldots + (\lambda_m\mu_n b_m a_n) = 1$$

(b) ($\Rightarrow$) : When fg is primitive. We prove by contradiction and suppose g is not primitive. Since fg is primitive, we may assume

$$b_0 a_0 \xi_0 + (b_1 a_0 + b_0 a_1)\xi_1 + (b_2 a_0 + b_1 a_1 + b_0 a_2)\xi_2 + \ldots + b_m a_n \xi_{m+n} = 1$$

Therefore

$$\begin{aligned}&(a_0\xi_0 + a_1\xi_1 + \ldots + a_n\xi_n)b_0\\ &+(a_0\xi_1 + a_1\xi_2 + \ldots + a_n\xi_{n+1})b_1\\ &+(a_0\xi_2 + a_1\xi_3 + \ldots + a_n\xi_{n+2})b_2\\ &\vdots\\ &+(a_0\xi_m + a_1\xi_{m+1} + \ldots + a_n + \xi_{m+n})b_m = 1\end{aligned}$$

Contradictory.

Remark 1

Consider the sum of terms in each of $n+m-1$ anti-diagonals

$$\begin{array}{c} \\ \mu_0 \\ \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{array} \begin{array}{c} \begin{array}{ccccc} \lambda_0 & \lambda_1 & \lambda_2 & \cdots & \lambda_m \end{array} \\ \begin{bmatrix} b_0a_0 & b_1a_0x & b_2a_0x^2 & \cdots & b_ma_0x^m \\ b_0a_1x & b_1a_1x^2 & b_2a_1x^3 & \cdots & b_ma_1x^{m+1} \\ b_0a_2x^2 & b_1a_2x^3 & b_2a_2x^4 & \cdots & b_ma_2x^{m+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_0a_nx^n & b_1a_nx^{n+1} & b_2a_nx^{n+2} & \cdots & b_ma_nx^{m+n} \end{bmatrix} \end{array}$$

Exercise 3

Generalize the results of Exercise 2 to a polynomial ring $A[x_1, \ldots, x_r]$ in several indeterminates.

Solution. We can prove it by induction. View $A[x_1, \ldots, x_n]$ as a ring over $A[x_1, \ldots, x_{n-1}]$ in an indeterminate x_n.

Exercise 4

In the ring $A[x]$, the Jacobson radical is equal to the nilradical.

Solution. Recall that the Jacobson radical is the intersection of all maximal ideals, while the nilradical is the intersection of all prime ideals. We'll prove that every prime ideal $p \subseteq A[x]$ is maximal:

$$\forall \text{prime ideal } p, \text{ it's maximal}$$

$\Leftrightarrow$

$$\forall \text{ prime ideal } p, \text{ Krull dimension of } p \text{ is } 0$$

$\Leftrightarrow$

$$\dim A[x] = 0$$

$\Leftrightarrow$(By *[Algebraic Geometry - Hartshorne]Proposition 1.7*)

$$\dim \{\emptyset\} = 0, \ \{\emptyset\} \text{ in affine 1-space } A$$

Which is trivial.

Remark 2

By induction, we'll get Jacobson radical being equal to nilradical in $A[x_1, x_2, \ldots, x_n], \forall n \in \mathbb{Z}^+$

Exercise 5

Let A be a ring and let $A[[x]]$ be the ring of formal power series $f = \sum_{n=0}^{\infty} a_n x^n$ with coefficients in A. Show that

(i) f is a unit in $A[[x]] \Leftrightarrow a_0$ is a unit in A.

(ii) If f is nilpotent, then a_n is nilpotent for all $n \geq 0$. Is the converse true? (See Chapter 7, Exercise 2.)

(iii) f belongs to the Jacobson radical of $A[[x]] \Leftrightarrow a_0$ belongs to the Jacobson radical of A.

(iv) The contraction of a maximal ideal $\mathfrak{m}$ of $A[[x]]$ is a maximal ideal of A, and $\mathfrak{m}$ is generated by $\mathfrak{m}^c$ and x.

(v) Every prime ideal of A is the contraction of a prime ideal of $A[[x]]$.

Solution. (i) (a) ($\Rightarrow$) : Suppose $fg = 1$, where

$$g = \sum_{n=0}^{\infty} b_n x^n$$

By checking the constant term of fg, we get

$$a_0 b_0 = 1$$

Thus a_0 is a unit in A.

(b) ($\Leftarrow$) : Notice

$$(a_0 + a_1 x + \ldots)(a_0^{-1} - a_1 a_0^{-2} x + (a_1^2 a_0^{-3} - a_2 a_0^{-2}) + \ldots) = 1$$

Such construction of f^{-1} can goes forever, so f is a unit in $A[[x]]$.

(ii) (a) When f is nilpotent, obviously a_0 is nilpotent. Nilpotent elements closed under addition, so $f - a_0$ is nilpotnet, indicating a_1 being nilpotent. Iterate the process above and we get $a_0, a_1, a_2, \ldots$ being nilpotent.

(b) No. Counterexample can be reached on *MathStackExchange*.

(iii) (a) ($\Leftarrow$) : For a_0 in maximal ideal I, we can easily check that

$$\mathfrak{I}_I = \left\{ b_0 + b_1 x + b_2 x^2 + \ldots \mid b_0 \in I, b_1, b_2, \ldots \in A \right\}$$

Is a maximal ideal in $A[[x]]$. Therefore $f = \sum_{n=0}^{\infty} a_n x^n \in \mathfrak{I}_I$.
So when a_0 is in the Jacobson Radical of A, f must be in the Jacobson Radical of $A[[x]]$.

(b) ($\Rightarrow$) : We only need to prove the maximal ideals of $A[[x]]$ are all in the form

$$\mathfrak{I}_I = \left\{ b_0 + b_1 x + b_2 x^2 + \ldots \mid b_0 \in I, b_1, b_2, \ldots \in A \right\}$$

where I is a maximal ideal of A.
Indeed, it's easy to see that the constant terms of all the elements in a arbitary ideal $\mathfrak{I}$ of $A[[x]]$ forms an ideal in A, we denote then by I and let $\mathfrak{I}_I = \mathfrak{I}$. When $\mathfrak{I}_I$ is maximal, I must be maximal in A, otherwise $\mathfrak{I}_I$ can expand to a larger ideal $\mathfrak{I}_{\overline{I}}$ where $\overline{I}$ is a maximal ideal containing I.
Hence the ideal of $A[[x]]$ is maximal if and only if the constant term ideal is maximal in A. And f is in a arbitary maximal ideal will indicate a_0 belongs to the correspond maximal ideal. Deduction of Jacobson Radical follows immediately.

(iv) Proved in (*iii*) above.

(v) **(UNFINISHED)** *(I'm not sure whether I understood this problem correctly.)*
Consider

$$\begin{aligned} \varphi : A &\longrightarrow A[[x]] \\ a &\longrightarrow a + 0x + 0x^2 + \ldots \end{aligned}$$

And define φ^{-1} by taking the constant term in the formal power series. For prime ideal $\mathfrak{p} \subseteq A$,

$$\mathfrak{p}A[[x]] = \left\{ a_0 + a_1 x + a_2 x^2 + \ldots \mid a_0, a_1, \ldots \in \mathfrak{p} \right\}$$

We can prove it's a

Remark 3
*The proof in **Ex.1.5(ii)** is identical to the one in **Ex.1.2(ii)***

Exercise 6

A ring A is such that every ideal not contained in the nilradical contains a nonzero idempotent (that is, an element e such that $e^2 = e \neq 0$). Prove that the nilradical and Jacobson radical of A are equal.

Solution. **(UNFINISHED)**

By *Proposition 1.11*, if the nilradical is equal to the Jacobson radical, since any prime ideal would contain the Jacobson radical, i.e. intersection of all maximal ideals, it would contain some of those maximal ideals.

Therefore the problem is equivalent to prove that every prime ideals in A is maximal,

Exercise 7

Let A be a ring in which every element x satisfies $x^n = x$ for some $n > 1$ (depending on x). Show that every prime ideal in A is maximal.

Solution. For any prime ideal $\mathfrak{p} \subseteq A$, consider the map

$$A \longrightarrow A/\mathfrak{p}$$

From the problem we know $\forall x \in A, \exists n > 0, \text{s.t. } x^n = x$. So $x(x^{n-1} - 1) = 0$, and we obtain

$$\overline{x}(\overline{x}^{n-1} - \overline{1}) = 0$$

We may assume $x \notin \mathfrak{p}$, i.e. $\overline{x} \neq 0$. Since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is a integral domain. So $\overline{x}^{n-1} = \overline{1}$, $\overline{x}$ has its inverse in $A/\mathfrak{p}$. Because x was an arbitary element in $A\backslash\mathfrak{p}$, $A/\mathfrak{p}$ is a field. Hence $\mathfrak{p}$ is maximal.

Exercise 8

Let A be a ring $\neq 0$. Show that the set of prime ideals of A has minimal elements with respect to inclusion.

Solution. Consider the intersection of all prime ideals of A, which is the nilradical and obviously minimal with respect to inclusion.

Exercise 9

Let $\mathfrak{a}$ be an ideal $\neq (1)$ in a ring A. Show that $\mathfrak{a} = \mathrm{Rad}(\mathfrak{a}) \Leftrightarrow \mathfrak{a}$ is an intersection of prime ideals.

Solution. Recall that $\mathrm{Rad}(\mathfrak{a})$ is the intersection of all prime ideals containing $\mathfrak{a}$. The proof can be obtained directly from it.

Exercise 10

Let A be a ring, $\mathfrak{N}$ its nilradical. Show that the following are equivalent:

(i) A has exactly one prime ideal;

(ii) every element of A is either a unit or nilpotent;

(iii) $A/\mathfrak{N}$ is a field.

Solution. (a) $(i) \Rightarrow (ii)$: If A has exactly one prime ideal $\mathfrak{p}$, such ideal will be the unique maximal ideal of A.

Recall that, by *Corollary 1.5*, every non-unit of A lies in $\mathfrak{p}$, therefore nilpotent. On the other hand, others will be the uni of A.

(b) $(ii) \Rightarrow (iii)$: $\forall$non-zero $\overline{a} \in \overline{A}, \exists a^{-1} \in A\backslash\mathfrak{R}, \text{s.t. } aa^{-1} = 1$. And we have $\overline{a}\overline{a^{-1}} = \overline{1}$.

(c) $(iii) \Rightarrow (i)$: $A/\mathfrak{R}$ is a field if and only if $\mathfrak{R}$ is a maximal ideal. Because a nilradical is the intersection of all prime ideals, we can see $\mathfrak{p}$ is unique and maximal in A.

Exercise 11

A ring A is Boolean if $x^2 = x$ for all $x \in A$. In a Boolean ring A, show that

(i) $2x = 0$ for all $x \in A$;

(ii) every prime ideal $\mathfrak{p}$ is maximal, and $A/\mathfrak{p}$ is a field with two elements;

(iii) every finitely generated ideal in A is principal.

Solution. (i) $2x = 4x^2 = 4x$, so $2x = 0$.

(ii) For any prime ideal $\mathfrak{p}$, consider the map

$$\varphi : A \longrightarrow A/\mathfrak{p}$$
$$x \longrightarrow \overline{x}$$

Since $x^2 = x$, $\overline{x}^2 = \overline{x}, \overline{x}(\overline{x} - 1) = 0$. $A/\mathfrak{p}$ is a integral domain, so $\overline{x} = 0$ or $\overline{x} = 1$. Hence $A/\mathfrak{p}$ only contains two elements, i.e. $\overline{1}$ and $\overline{0}$, and it's a field for sure.

(iii) For $I = (x, y)$, consider $z = x + y + xy$, we'll prove $(z) = (x, y)$.

By definition, $(z) \subseteq (x, y)$. Also, notice

$$zx = (x + y + xy)x = x + xy + xy = x$$

So $x \in (z)$. In the same way we have $y \in (z)$, thus $(z) = (x, y)$. Rest of the proof can be done by induction.

Exercise 12

A local ring contains no idempotent $\neq 0, 1$.

Solution. Suppse we have a nontrivial idempotent e. By $e^2 = e$ we have

$$e(1 - e) = 0$$

So both e and $1 - e$ is a zero-divisor, hence they're not unit and lies in the unique maximal ideal. Therefore $1 = e + (1 - e)$ is in the maximal ideal, contradictory.

Construction of an algebraic closure of a field (E. Artin)

Exercise 13

Let K be a field and let Σ be the set of all irreducible monic polynomials f in one indeterminate with coefficients in K. Let A be the polynomial ring over K generated by indeterminates x_f, one for each $f \in \Sigma$. Let $\mathfrak{a}$ be the ideal of A generated by the polynomials $f(x_f)$ for all $f \in \Sigma$. Show that $\mathfrak{a} \neq (1)$.

Let $\mathfrak{m}$ be a maximal ideal of A containing $\mathfrak{a}$, and let $K_1 = A/\mathfrak{m}$. Then K_1 is an extension field of K in which each $f \in \Sigma$ has a root. Repeat the construction with K_1 in place of K, obtaining a field K_2, and so on. Let $L = \bigcup_{n=1}^{\infty} K_n$. Then L is a field in which each $f \in \Sigma$ splits completely into linear factors. Let $\bar{K}$ be the set of all elements of L which are algebraic over K. Then $\bar{K}$ is an algebraic closure of K.

Solution. **(POOR-WRITTEN UNCLEAR PROOF)**

We prove by contradiction: Suppose $\mathfrak{a} = (1)$, then $\exists$polynomial $F \in A[f(x_f)]$, for all f and x_f, such that

$$F(f(x_f), \ldots) = 1, \text{ finitely many } f \in \Sigma$$

For those F, we pick out one with the least number of $f \in \Sigma$ being involved.

For a specific $f_0 \in \Sigma$ which was involved in the polynomial F above, consider the terms of f_0 with its highest degree α_0 in $F(f(x_f), \ldots)$. Because $F(f(x_f), \ldots) = 1$, those terms with degree α_0 sums up to 0, i.e.

$$f(x_f)^{\alpha_0}(\text{rest of the terms}) = 0$$

Apparently $f(x_f)^{\alpha_0} \neq 0$, so the rest of the terms sums up to 0. Then for the rest of the terms, we can again similarly pick out a f_1, α_1 and sort out "rest of the terms" to be 0. We iterate the process above. Since there're only finitely many terms, the process must stop, making a contradiction.

(We don't need to worry about characteristic of K is finite or not, since w.l.o.g. we can let those terms in F which adds up over the characteristic vanish in the first place.)

Exercise 14

In a ring A, let Σ be the set of all ideals in which every element is a zero-divisor. Show that the set Σ has maximal elements and that every maximal element of Σ is a prime ideal. Hence the set of zero-divisors in A is a union of prime ideals.

Solution. (i) By Zorn's lemma, Σ has maximal elements.

(ii) $\forall$maximal $I \subseteq \Sigma$, suppose $a, b \in A$, s.t. $ab \in I$. I consists of zero-divisors, so $\exists u \in I$, s.t. $(ab)u = 0$. Then

$$a(bu) = 0$$

indicating a being a zero-divisor, so does b. Since I is maximal, $a \in I$ (or $b \in I$)

(iii) This can be obtained immediately by taking the union of every maximal element in Σ

The prime spectrum of a ring

Exercise 15

Let A be a ring and let X be the set of all prime ideals of A. For each subset E of A, let $V(E)$ denote the set of all prime ideals of A which contain E. Prove that

(i) if $\mathfrak{a}$ is the ideal generated by E, then $V(E) = V(\mathfrak{a}) = V(\mathrm{Rad}(\mathfrak{a}))$.

(ii) $V(0) = X$, $V(1) = \varnothing$.

(iii) if $(E_i)_{i \in I}$ is any family of subsets of A, then

$$V\Big(\bigcup_{i \in I} E_i\Big) = \bigcap_{i \in I} V(E_i).$$

(iv) $V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$ for any ideals $\mathfrak{a}, \mathfrak{b}$ of A.

These results show that the sets $V(E)$ satisfy the axioms for closed sets in a topological space. The resulting topology is called the Zariski topology. The topological space X is called the prime spectrum of A, and is written $\mathrm{Spec}(A)$.

Solution. Trivial. *(See [Abstract Algebra - D&F(3rd Edition)]Chap 12, [Algebraic Geometry - Hartshorne]Sect 1)*

Exercise 16

Draw pictures of $\mathrm{Spec}(\mathbb{Z})$, $\mathrm{Spec}(\mathbb{R})$, $\mathrm{Spec}(\mathbb{C}[x])$, $\mathrm{Spec}(\mathbb{R}[x])$, $\mathrm{Spec}(\mathbb{Z}[x])$.

Solution. (i) $\mathrm{Spec}(\mathbb{Z}) = \{(p) \mid p \in \mathbb{Z} \text{ is prime } \}$

(ii) $\mathrm{Spec}(\mathbb{R}) = \{(0)\}$

(iii) $\mathrm{Spec}(\mathbb{C}[x]) = \{(p(x)) \mid p(x) \in \mathbb{C}[x], \deg p = 1\}$

(iv) $\mathrm{Spec}(\mathbb{R}[x]) = \{(p(x)) \mid p(x) \in \mathbb{R}[x]$ prime (or we can say irreducible) $\}$

(v) **(UNFINISHED)**
$\mathrm{Spec}(\mathbb{Z}[x]) = \{\}$

Exercise 17

For each $f \in A$, let X_f denote the complement of $V(f)$ in $X = \mathrm{Spec}(A)$. The sets X_f are open. Show that they form a basis of open sets for the Zariski topology, and that

(i) $X_f \cap X_g = X_{fg}$;

(ii) $X_f = \varnothing \Leftrightarrow f$ *is nilpotent;*

(iii) $X_f = X \Leftrightarrow f$ *is a unit;*

(iv) $X_f = X_g \Leftrightarrow \mathrm{Rad}((f)) = \mathrm{Rad}((g))$*;*

(v) X *is quasi-compact (that is, every open covering of* X *has a finite sub-covering).*

(vi) More generally, each X_f *is quasi-compact.*

(vii) An open subset of X *is quasi-compact if and only if it is a finite union of sets* X_f*.*

The sets X_f *are called basic open sets of* $X = \mathrm{Spec}(A)$*.*

[To prove (v), remark that it is enough to consider a covering of X *by basic open sets* X_{f_i} *(*$i \in I$*). Show that the* f_i *generate the unit ideal and hence that there is an equation of the form*

$$1 = \sum_{i \in J} g_i f_i \qquad (g_i \in A)$$

where J *is some finite subset of* I*. Then the* X_{f_i} *(*$i \in J$*) cover* X*.]*

Solution. **(UNFINISHED)** Obviously they form a basis of open sets for Zariski topology.

(i) Trivial.

(ii) Trivial.

(iii) Trivial.

(iv) Trivial.

(v) Without loss of generation ww may only conside X convered by basic open sets X_{f_i}, since every general open sets can be in the form of union of basis.

Now

$$\forall \bigcup_{k \in I} X_{f_k} \supseteq X = \mathrm{Spec}(A) \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcup_{i=1}^{n} X_{f_{k_i}} \supseteq X$$

$\Leftrightarrow$

$$\forall \bigcap_{k \in I} V(f_k) \subseteq \emptyset \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcap_{i=1}^{n} V(f_{k_i}) \subseteq \emptyset$$

$\Leftrightarrow$

$$\forall \bigcap_{k \in I} V(f_k) = \emptyset \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcap_{i=1}^{n} V(f_{k_i}) = \emptyset$$

$\Leftrightarrow$

$$\forall (f_1, f_2, \ldots) = (1) \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } (f_{k_1}, \ldots, f_{k_n}) = (1)$$

When $\forall (f_1, f_2, \ldots) = (1)$, $\exists g_{k_1}, \ldots, g_{k_n} \in A, \text{s.t. } 1 = \sum a_{k_i} f_{k_i}, a \in A$, which implies $(1) = (f_{k_1}, \ldots, f_{k_n})$.

(vi) Similarly,

$$\forall \bigcup_{k \in I} X_{f_k} \supseteq X_f \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcup_{i=1}^{n} X_{f_{k_i}} \supseteq X_f$$

$\Leftrightarrow$

$$\forall \bigcap_{k \in I} V(f_k) \subseteq V(f) \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcap_{i=1}^{n} V(f_{k_i}) \subseteq V(f)$$

$\Leftrightarrow$

$$\forall f_1, f_2, \ldots \text{s.t. } f \in (f_1, f_2, \ldots) \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } f = a_1 f_1 + \ldots + a_n f_n, (a_k \in A)$$

Which is trivial.

(vii) Since $\{X_f\}$ forms a basis for $\mathrm{Spec}(A)$, $\exists$famliy$\mathcal{F}$, s.t. $X_0 = \bigcup_{f\in\mathcal{F}} X_f$. Now

$$X_0 \text{ quasi compact}$$

$$\Leftrightarrow$$

$$\forall X_{f_1}, X_{f_2}, \ldots \in A, \text{s.t. } \bigcup_{k=1}^{\infty} X_{f_k} \supseteq \bigcup_{f\in\mathcal{F}} X_f, \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcup_{i=1}^{n} X_{f_{k_i}} \supseteq \bigcup_{f\in\mathcal{F}} X_f$$

$$\Leftrightarrow$$

$$\forall f_1, f_2, \ldots, \text{s.t. } \bigcap_{k=1}^{\infty} V(f_k) \subseteq \bigcap_{f\in\mathcal{F}} V(f), \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcap_{i=1}^{n} V(f_k) \subseteq \bigcap_{f\in\mathcal{F}} V(f)$$

$$\Leftrightarrow$$

$$\forall f_1, f_2, \ldots \in A, \text{s.t. } \mathcal{F} \subseteq (f_1, f_2, \ldots), \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \mathcal{F} \subseteq (f_{k_1}, \ldots, f_{k_n})$$

(i) ($\Rightarrow$) : We prove by contradiction and suppose $\mathcal{F}$ can only be infinite, then let $f_1, f_2 \ldots$ be the elements in $\mathcal{F}$, and such $f_{k_1}, \ldots, f_{k_n}$ cannot exist.

(ii) ($\Leftarrow$) : When $\mathcal{F}$ is finite, it is a direct proposition from *(vi)*.

Exercise 18

For psychological reasons it is sometimes convenient to denote a prime ideal of A by a letter such as x or y when thinking of it as a point of $X = \mathrm{Spec}(A)$. When thinking of x as a prime ideal of A, we denote it by $\mathfrak{p}_x$ (logically, of course, it is the same thing). Show that

(i) the set $\{x\}$ is closed (we say that x is a "closed point") in $\mathrm{Spec}(A) \Leftrightarrow \mathfrak{p}_x$ is maximal;

(ii) $\overline{\{x\}} = V(\mathfrak{p}_x)$;

(iii) $y \in \overline{\{x\}} \Leftrightarrow \mathfrak{p}_x \subseteq \mathfrak{p}_y$;

(iv) X is a T_0-space (this means that if x, y are distinct points of X, then either there is a neighborhood of x which does not contain y, or else there is a neighborhood of y which does not contain x).

Solution. (i) $\{x\}$ is closed if and only if the only prime ideal containing $\mathfrak{p}_x$ is $\mathfrak{p}_x$ itself, which is equivalent to $\mathfrak{p}_x$ being maximal.

(ii) Trivial.

(iii) $y \in \{\overline{x}\} \Leftrightarrow \mathfrak{p}_y \in V(\mathfrak{p}_x) \Leftrightarrow \mathfrak{p}_x \subseteq \mathfrak{p}_y$.

(iv) *(What's the definition of "neighborhood" here? I'll interpret it as "closed neighborhood" anyway.)*

Consider $V(x), V(y)$, which is a neighborhood of x, y respectively. Soppose $x \in V(\mathfrak{p}_y), y \in V(\mathfrak{p}_x)$, then by *(iii)* we get $\mathfrak{p}_x = \mathfrak{p}_y$, contradictory.

Exercise 19

A topological space X is said to be irreducible if $X \neq \varnothing$ and if every pair of non-empty open sets in X intersect, or equivalently if every non-empty open set is dense in X. Show that $\mathrm{Spec}(A)$ *is irreducible if and only if the nilradical of A is a prime ideal.*

Solution. (Beautiful!)

In ring A, we consider the category $\mathscr{C}$ of ideals, morphisms umder inclusions. We take its subcategory $\mathscr{C}_0$ of prime ideals, i.e.$\mathrm{Obj}(\mathscr{C}_0) = \mathrm{Spec}(A)$.

In this case, the initial object of $\mathscr{C}$ is the nilradial of A, and it lies in $\mathscr{C}_0$ if and only if the nilradical is prime. Therefore, $\mathscr{C}_0$ has a initial object, i.e. the partially ordered set of $\mathrm{Spec}(A)$ consists of a connected tree instead of a union of disjoint trees, if and only if $\mathrm{Nil}(A)$ is prime.

On the other hand, for any two subsets of A, denoted by E, F, we denote the complement of $V(E), V(F)$ by X_E, X_F respectively. We can have

$$X_E \cap X_F \neq \emptyset$$

$\Leftrightarrow$

$$V(E) \cup V(F) \neq \mathrm{Spec}(A)$$

$\Leftrightarrow$

$$\mathrm{Nil}(A) \subseteq \mathrm{Spec}(A)$$

(Becuase $V(\mathrm{Nil}(A)) = \mathrm{Spec}(A) = X$. Otherwise when nilradical is not prime, $\mathscr{C}_0$ consists of distinct trees with each root $r_1, r_2, \ldots$, from which we can let $E = \{r_1\}, F = \{r_2, \ldots\}$ and have $V(E) \cup V(F) = X$.)

Exercise 20

Let X be a topological space.

(i) If Y is an irreducible (Exercise 19) subspace of X, then the closure $\bar{Y}$ of Y in X is irreducible.

(ii) Every irreducible subspace of X is contained in a maximal irreducible subspace.

(iii) The maximal irreducible subspaces of X are closed and cover X. They are called the irreducible components of X. What are the irreducible components of a Hausdorff space?

(iv) If A is a ring and $X = \mathrm{Spec}(A)$, then the irreducible components of X are the closed sets $V(\mathfrak{p})$, where $\mathfrak{p}$ is a minimal prime ideal of A (Exercise 8).

Solution. **(SKIPPED)** *(I haven't learnt topology yet.)*

Exercise 21

Let $\phi : A \to B$ be a ring homomorphism. Let $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$. If $\mathfrak{q} \in Y$, then $\phi^{-1}(\mathfrak{q})$ is a prime ideal of A, i.e., a point of X. Hence ϕ induces a mapping $\phi^* : Y \to X$. Show that

(i) If $f \in A$ then $\phi^{*-1}(X_f) = Y_{\phi(f)}$, and hence that ϕ^* is continuous.

(ii) If $\mathfrak{a}$ is an ideal of A, then $\phi^{*-1}(V(\mathfrak{a})) = V(\mathfrak{a}^e)$.

(iii) If $\mathfrak{b}$ is an ideal of B, then $\overline{\phi^*(V(\mathfrak{b}))} = V(\mathfrak{b}^c)$.

(iv) If ϕ is surjective, then ϕ^* is a homeomorphism of Y onto the closed subset $V(\mathrm{Ker}(\phi))$ of X. (In particular, $\mathrm{Spec}(A)$ and $\mathrm{Spec}(A/\mathfrak{N})$ (where $\mathfrak{N}$ is the nilradical of A) are naturally homeomorphic.)

(v) If ϕ is injective, then $\phi^*(Y)$ is dense in X. More precisely, $\phi^*(Y)$ is dense in $X \Leftrightarrow \mathrm{Ker}(\phi) \subseteq \mathfrak{N}$.

(vi) Let $\psi : B \to C$ be another ring homomorphism. Then $(\psi \circ \phi)^* = \phi^* \circ \psi^*$.

(vii) Let A be an integral domain with just one non-zero prime ideal $\mathfrak{p}$, and let K be the field of fractions of A. Let $B = (A/\mathfrak{p}) \times K$. Define $\phi : A \to B$ by $\phi(x) = (\bar{x}, x)$, where $\bar{x}$ is the image of x in $A/\mathfrak{p}$. Show that ϕ^* is bijective but not a homeomorphism.

Solution. **(UNFINISHED)**

$$\begin{array}{ccc} A & \xrightarrow{\quad\phi\quad} & B \\ \downarrow{\scriptstyle V()} & & \downarrow{\scriptstyle V()} \\ X = \mathrm{Spec}(A) & \xleftarrow[\phi^*]{} & Y = \mathrm{Spec}(B) \end{array}$$

(i) For $\mathfrak{q} \in Y_{\phi(f)}$,

$$\mathfrak{q} \in \phi^{*-1}(X_f)$$

$\Leftrightarrow$

$$\phi^*(\mathfrak{q}) \in X_f$$

$\Leftrightarrow$

$$\phi^*\mathfrak{q} \notin V(f)$$

$\Leftrightarrow$

$$f \notin \phi^*(\mathfrak{q}) = \phi^{-1}(\mathfrak{q})$$

$\Leftrightarrow$

$$\phi(f) \notin \mathfrak{q}$$

$\Leftrightarrow$

$$\mathfrak{q} \in Y_{\phi_{(f)}}$$

So $\phi^{*-1}(X_f) = Y_{\phi(f)}$.

(ii) For any prime ideal $\mathfrak{q} \in Y$,

$$\mathfrak{q} \in \phi^{*-1}(V(\mathfrak{a}))$$

$\Leftrightarrow$

$$\phi^*(\mathfrak{q}) \in V(\mathfrak{a})$$

$\Leftrightarrow$

$$\mathfrak{a} \subseteq \phi^*(\mathfrak{q})$$

$\Leftrightarrow$

$$\phi(\mathfrak{a}) \subseteq \phi(\phi^*(\mathfrak{q})) = \mathfrak{q}$$

$\Leftrightarrow$

$$\mathfrak{q} \in V(\phi(\mathfrak{a})) = V(\mathfrak{a}^e)$$

So $\phi^{*-1}(V(\mathfrak{a})) = V(\mathfrak{a}^e)$.

(iii) $\forall \mathfrak{q} \in V(\mathfrak{b})$

$$\phi^{-1}(\mathfrak{b}) \subseteq \phi^*(\mathfrak{q})$$

$\Leftrightarrow$

$$\mathfrak{b} \subseteq \mathfrak{q}$$

Which is true by definition, so $\overline{\phi^*(V(\mathfrak{b}))} \subseteq V(\phi^{-1}(\mathfrak{b}))$. **(UNFINISHED)**

(iv) (a) From *(i)* we already know ϕ^* is continuous. Since we now have ϕ being surjection, ϕ^* is obviously a bijection. Also, the continuity of derives directly from *(iii)*.

(b) In particular, let $Y = A/\mathfrak{R}$ and ϕ be the canonical homomorphism.

(v) We directly prove the "More precisely" part:

$$\phi^*(Y) \text{ dense in } X$$

$\Leftrightarrow$

$$\overline{\phi^*(Y)} = X \tag{1}$$

$\Leftrightarrow$

$$\overline{\phi^*(V(\mathrm{Nil}(B)))} = X$$

$\Leftrightarrow$

$$\overline{\phi^*(V(\mathrm{Nil}(B)))} = V(\mathrm{Nil}(A))$$

$\Leftrightarrow$ (by *(iii)*)

$$V(\mathrm{Nil}(B)^c) = V(\mathrm{Nil}(A))$$

$\Leftrightarrow$

$$V(\phi^{-1}(\mathrm{Nil}(B))) = V(\mathrm{Nil}(A))$$

$\Leftrightarrow$

$$\mathrm{Rad}(\phi^{-1}(\mathrm{Nil}(B))) = \mathrm{Nil}(A) \tag{2}$$

To prove ((1) $\Leftrightarrow$ (2)), we only need

$$\mathrm{Ker}(\phi) \subseteq \mathrm{Rad}(\phi^{-1}(\mathrm{Nil}(B)))$$

$\Leftarrow$

$$\mathrm{Ker}(\phi) \subseteq \phi^{-1}(\mathrm{Nil}(B))$$

$\Leftrightarrow$

$$0_B \subseteq \mathrm{Nil}(B)$$

Which is trivial.

(vi) Trivial.

(vii)

Exercise 22

Let $A = \prod_{i=1}^{n} A_i$ be the direct product of rings A_i. Show that $\mathrm{Spec}(A)$ *is the disjoint union of open (and closed) subspaces X_i, where X_i is canonically homeomorphic with* $\mathrm{Spec}(A_i)$.

Conversely, let A be any ring. Show that the following statements are equivalent:

(i) $X = \mathrm{Spec}(A)$ *is disconnected.*

(ii) $A \cong A_1 \times A_2$ *where neither of the rings A_1, A_2 is the zero ring.*

(iii) A contains an idempotent $\neq 0, 1$.

In particular, the spectrum of a local ring is always connected (Exercise 12).

Solution. (a) $\forall k \in [1, n] \cap Z$,

$$X_k = \{A_1 \times \ldots \times A_{k-1} \times \mathfrak{p} \times A_{k+1} \times \ldots \times A_n \mid \mathfrak{p} \in \mathrm{Spec}(A_k)\}$$

So we can define a map

$$\varphi : \mathrm{Spec}(A_k) \longrightarrow X_k$$

by

$$\mathfrak{p} \longmapsto A_1 \times \ldots \times A_{k-1} \times \mathfrak{p} \times A_{k+1} \times \ldots \times A_n$$

and the homeomorphic property is trivial in the Zariski topology defined in this way.

(b) (I) $(ii) \Rightarrow (i)$: From *(a)* we see

$$X_1 = \{\mathfrak{p} \times A_2 \mid \mathfrak{p} \in \mathrm{Spec}(A_1)\}, X_2 = \{A_1 \times \mathfrak{p} \mid \mathfrak{p} \in \mathrm{Spec}(A_2)\}$$

So $X_1 \cap X_2 = \emptyset$.

(II) $(iii) \Rightarrow (ii)$: Suppose we have $e \in A$, s.t. $e^2 = 2, e \notin \{0, 1\}$. Then $A \cong eA \times (1 - e)A$.

(III) $(i) \Rightarrow (iii)$:

$$X \text{ disconnect}$$

$$\Leftrightarrow$$

$$X \text{ is a union of disjoint nontrivial clopen set}$$

$$\Leftrightarrow$$

$$\exists \text{ideal } I, J, \text{s.t. } V(I) \cap V(J) = V(I + J) = \emptyset, V(I) \cup V(J) = V(IJ) = V(I \cap J) = X$$

When such I, J exist, let $a \in I, b \in J$, s.t. $a + b = 1$. Since $V(IJ) = X, IJ \subseteq \mathrm{Nil}(A)$, so $\exists n \in \mathbb{Z}^+$, s.t. $(ab)^n = 0$. Consider

$$s_1 = \mathrm{C}_{2n}^0 a^{2n} + \mathrm{C}_{2n}^1 a^{2n-1} b + \ldots + \mathrm{C}_{2n}^{n-1} a^{n+1} b^{n-1}$$

$$s_2 = \mathrm{C}_{2n}^{2n} b^{2n} + \mathrm{C}_{2n}^{2n-1} a b^{2n-1} + \ldots + \mathrm{C}_{2n}^{n+1} a^{n-1} b^{n+1}$$

Now

$$s_1 + s_2 = s_1 + s_2 + \mathrm{C}_{2n}^{n} a^n b^n = (a + b)^{2n} = 1$$

and

$$s_1 s_2 = 0$$

So

$$s_1(1 - s_1) = 0$$

s_1, s_2 are nontrivial idempotent.

Exercise 23

Let A be a Boolean ring (Exercise 11), and let $X = \mathrm{Spec}(A)$.

(i) For each $f \in A$, the set X_f (Exercise 17) is both open and closed in X.

(ii) Let $f_1, \ldots, f_n \in A$. Show that $X_{f_1} \cup \cdots \cup X_{f_n} = X_f$ for some $f \in A$.

(iii) The sets X_f are the only subsets of X which are both open and closed. [Let $Y \subseteq X$ be both open and closed. Since Y is open, it is a union of basic open sets X_f. Since Y is closed and X is quasi-compact (Exercise 17), Y is quasi-compact. Hence Y is a finite union of basic open sets; now use (ii) above.]

(iv) X is a compact Hausdorff space.

Solution. (i) X_f is open by definition, and we only have to prove that X_f is closed.

For any nonzero, non-unital $f \in A$ consider $1 - f$. Notice:

$$\begin{aligned} V(f) \cap V(1-f) &= V((f) + (1-f)) \\ &= V(1) \\ &= \emptyset \end{aligned}$$

$$\begin{aligned} V(f) \cup V(1-f) &= V((f)(1-f)) \\ &= V(0) \\ &= X \end{aligned}$$

Therefore $V(f) = X_{1-f}$.

(ii) Because

$$\bigcap_{k=1}^{n} V(f_k) = V\left(\sum_{k=1}^{n} (f_k)\right)$$

where the ideal $\sum_{k=1}^{n} (f_k)$ is finitely generated, therefore principal by *Ex-1.11(iii)*. So $\exists f \in A$, s.t.

$$(f) = (f_1, f_2, \ldots., f_n) = \sum_{k=1}^{n} (f_k)$$

and therefore

$$V(f) = \bigcap_{k=1}^{n} V(f_k)$$

(iii) For any clopen subset $Y \subseteq X$, let $Z = X \backslash Y$. By definition Z is also clopen. From *Prop-1.11(i)*,

$$\forall \mathfrak{p} \in Y, \mathfrak{p} \backslash \left(\bigcup_{\mathfrak{q} \in Y \backslash \mathfrak{p}} \mathfrak{q} \right)$$

is nonempty. Thus we can pick out $f \in \mathfrak{p} \backslash \left(\bigcup_{\mathfrak{q} \in Y \backslash \mathfrak{p}} \mathfrak{q} \right)$ and consider $(1 - f)$: From *(i)* we know $V(f), V(1-f)$ is a partition of X, so by $f \in \mathfrak{p}, \mathfrak{p} \in Y$ we have

$$V(f) = Y, V(1-f) = Z$$

But since $f \notin \bigcup_{\mathfrak{q} \in Y} \mathfrak{q}$, $V(f) \subsetneq Y$, contradictory.

(iv) X is compact by *Ex-17(iv)*.

To prove X is Hausdorff, we need to prove for $\forall \mathfrak{p}, \mathfrak{q} \in X, \mathfrak{p} \neq \mathfrak{q}$,

$$\exists \text{open set } X_1, X_2, \text{s.t. } \mathfrak{p} \in X_1, \mathfrak{q} \in X_2, X_1 \cap X_2 = \emptyset$$

$\Leftrightarrow$

$$\exists \text{closed set } V_1, V_2, \text{s.t. } \mathfrak{p} \notin V_1, \mathfrak{q} \notin V_2, V_1 \cup V_2 = X$$

$\Leftrightarrow$

$$\exists \text{closed set } V_1, V_2, \text{s.t. } \mathfrak{p} \in V_2 \backslash V_1, \mathfrak{q} \in V_1 \backslash V_2, V_1 \cup V_2 = X$$

Such V_1, V_2 can be obtained by

$$V_1 = X \backslash \{\mathfrak{q}\}, V_2 = X \backslash \{\mathfrak{p}\}$$

Exercise 24

Let L be a lattice, in which the sup and inf of two elements a, b are denoted by $a \vee b$ and $a \wedge b$ respectively. L is a Boolean lattice (or Boolean algebra) if

(i) L has a least element and a greatest element (denoted by $0, 1$ respectively).

(ii) Each of $\vee, \wedge$ is distributive over the other.

(iii) Each $a \in L$ has a unique "complement" $a' \in L$ such that $a \vee a' = 1$ and $a \wedge a' = 0$.

(For example, the set of all subsets of a set, ordered by inclusion, is a Boolean lattice.)

Let L be a Boolean lattice. Define addition and multiplication in L by the rules

$$a + b = (a \wedge b') \vee (a' \wedge b), \qquad ab = a \wedge b.$$

Verify that in this way L becomes a Boolean ring, say $A(L)$.

Conversely, starting from a Boolean ring A, define an ordering on A as follows: $a \leq b$ means that $a = ab$. Show that, with respect to this ordering, A is a Boolean lattice. [The sup and inf are given by $a \vee b = a + b + ab$ and $a \wedge b = ab$, and the complement by $a' = 1 - a$.] In this way we obtain a one-to-one correspondence between (isomorphism classes of) Boolean rings and (isomorphism classes of) Boolean lattices.

Solution. (a) (I) For addition:

(i) $+$ is commutative because $\wedge, \vee$ are both commutative.

(ii)

$$\begin{aligned} a + 0 &= (a \wedge 1) \vee (a' \wedge 0) \\ &= a \vee 0 \\ &= a \end{aligned}$$

(iii)

$$\begin{aligned} a + a' &= (a \wedge a') \vee (a' \wedge a) \\ &= 0 \vee 0 \\ &= 0 \end{aligned}$$

(iv)

$$\begin{aligned} (a + b) + c &= (ab' \vee a'b) + c \\ &= (ab' \vee a'b)c \vee (ab + a'b')c \\ &= abc \vee a'bc \vee ab'c \vee abc' \end{aligned}$$

Which is symmetric with respect to a, b, c, therefore addition is associative.

(II) For multiplication:

(i)

$$ab = a \wedge b = b \wedge a = ba$$

(ii)

$$a \cdot 1 = a \wedge 1 = a$$

(iii) Associative of multiplication is given by the associativity of $\vee$.

(iv)

$$\begin{aligned} a(b+c) &= a(bc' \vee b'c) \\ &= abc' \vee ab'c \end{aligned}$$

$$\begin{aligned} ab + ac &= ((a' \vee b')ac) \vee (ab(a' \vee c')) \\ &= ab'c \vee abc' \end{aligned}$$

And we obtian the distributive law.

(III) For Boolean property: $a^2 = a \wedge a = a$.

(b) We'll prove one side, the other side cna be obtained by duality.

(i) Associativity:

$$\begin{aligned} a \vee (b \vee c) &= a \vee (b + c + bc) \\ &= a + b + c + bc + ab + ca + abc \end{aligned}$$

$$\begin{aligned} (a \vee b) \vee c &= (a + b + ab) \vee c \\ &= a + b + c + ab + ca + bc + abc \end{aligned}$$

So $(a \vee b) \vee c = a \vee (b \vee c)$.

(ii) Commutavity: Because multiplication is commutative.

(iii) Absorption:

$$\begin{aligned} a \vee (a \wedge b) &= a \vee (ab) \\ &= a + ab + aab \\ &= 3a \\ &= a \end{aligned}$$

(iv) Identity: $a \vee 0 = a$

(v) Distributivity:

$$\begin{aligned} a \vee (b \wedge c) &= a \vee (bc) \\ &= a + bc + abc \end{aligned}$$

$$\begin{aligned} (a \vee b) \wedge (a \vee c) &= (a + b + ab)(a + c + ac) \\ &= a + ac + ac + ab + bc + abc + ab + abc + abc \\ &= a + bc + abc \end{aligned}$$

So $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$.

(vi) Complements:

$$\begin{aligned} a \vee a' &= a + (1 - a) + a(1 - a) \\ &= 1 \end{aligned}$$

Exercise 25

From the last two exercises deduce Stone's theorem, that every Boolean lattice is isomorphic to the lattice of open-and-closed subsets of some compact Hausdorff topological space.

Solution. (Beautiful!)

Every Boolean lattice, namely L, is isomorphic to $Y = \{X_f \mid f \in A(L), X = \mathrm{Spec}(A(L))\}$

Exercise 26

Let A be a ring. The subspace of $\mathrm{Spec}(A)$ *consisting of the maximal ideals of A, with the induced topology, is called the maximal spectrum of A and is denoted by* $\mathrm{Max}(A)$*. For arbitrary commutative rings it does not have the nice functorial properties of* $\mathrm{Spec}(A)$ *(see Exercise 21), because the inverse image of a maximal ideal under a ring homomorphism need not be maximal.*

Let X be a compact Hausdorff space and let $C(X)$ denote the ring of all real-valued continuous functions on X (add and multiply functions by adding and multiplying their values). For each $x \in X$, let $\mathfrak{m}_x$ be the set of all $f \in C(X)$ such that $f(x) = 0$. The ideal $\mathfrak{m}_x$ is maximal, because it is the kernel of the (surjective) homomorphism $C(X) \to \mathbb{R}$ which takes f to $f(x)$. If $\tilde{X}$ denotes $\mathrm{Max}(C(X))$*, we have therefore defined a mapping $\mu : X \to \tilde{X}$, namely $x \mapsto \mathfrak{m}_x$.*

We shall show that μ is a homeomorphism of X onto $\tilde{X}$.

(i) Let $\mathfrak{m}$ be any maximal ideal of $C(X)$, and let $V = V(\mathfrak{m})$ be the set of common zeros of the functions in $\mathfrak{m}$: that is,

$$V = \{x \in X : f(x) = 0 \text{ for all } f \in \mathfrak{m}\}.$$

Suppose that V is empty. Then for each $x \in X$ there exists $f_x \in \mathfrak{m}$ such that $f_x(x) \neq 0$. Since f_x is continuous, there is an open neighborhood U_x of x in X on which f_x does not vanish. By compactness a finite number of the neighborhoods, say $U_{x_1}, \ldots, U_{x_n}$, cover X. Let

$$f = f_{x_1}^2 + \cdots + f_{x_n}^2.$$

Then f does not vanish at any point of X, hence is a unit in $C(X)$. But this contradicts $f \in \mathfrak{m}$, hence V is not empty.

Let x be a point of V. Then $\mathfrak{m} \subseteq \mathfrak{m}_x$, hence $\mathfrak{m} = \mathfrak{m}_x$ because $\mathfrak{m}$ is maximal. Hence μ is surjective.

(ii) By Urysohn's lemma (this is the only non-trivial fact required in the argument) the continuous functions separate the points of X. Hence $x \neq y \Rightarrow \mathfrak{m}_x \neq \mathfrak{m}_y$, and therefore μ is injective.

(iii) Let $f \in C(X)$; let

$$U_f = \{x \in X : f(x) \neq 0\}$$

and let

$$\tilde{U}_f = \{\mathfrak{m} \in \tilde{X} : f \notin \mathfrak{m}\}.$$

Show that $\mu(U_f) = \tilde{U}_f$. The open sets U_f (resp. $\tilde{U}_f$) form a basis of the topology of X (resp. $\tilde{X}$) and therefore μ is a homeomorphism.

Thus X can be reconstructed from the ring of functions $C(X)$.

Solution. (The hint is making the proble trivial enough.)

Form *(i),(ii)* we know μ is bijective.

On one hand, $\forall \mathfrak{m} \in \tilde{U}_f, \exists y \in X, \text{s.t. } \mathfrak{m} = \mathfrak{m}_y$. Since $\mathfrak{m}_y \in \tilde{U}_f, f \in \mathfrak{m}_y \Rightarrow f(y) \neq 0$. Therefore $\exists y \in U_f, \text{s.t. } \mu(y) = \mathfrak{m}_y$. Hence $\tilde{U}_f \subseteq \mu(U_f)$.

On the otherhand, $\forall x \in U_f$, we have $f(x) \neq 0$. Therefore $f \notin \mathfrak{m}_x \Rightarrow \exists \mathfrak{m}_x \in \tilde{U}_f, \text{s.t. } \mu(x) = \mathfrak{m}_x$. So $U_f \subseteq \tilde{U}_f$.

Affine algebraic varieties

Exercise 27

Let k be an algebraically closed field and let

$$f_\alpha(t_1, \ldots, t_n) = 0$$

be a set of polynomial equations in n variables with coefficients in k. The set X of all points $x = (x_1, \ldots, x_n) \in k^n$ which satisfy these equations is an affine algebraic variety.

Consider the set of all polynomials $g \in k[t_1, \ldots, t_n]$ with the property that $g(x) = 0$ for all $x \in X$. This set is an ideal $I(X)$ in the polynomial ring, and is called the ideal of the variety X. The quotient ring

$$P(X) = k[t_1, \ldots, t_n]/I(X)$$

is the ring of polynomial functions on X, because two polynomials g, h define the same polynomial function on X if and only if $g - h$ vanishes at every point of X, that is, if and only if $g - h \in I(X)$.

Let ξ_i be the image of t_i in $P(X)$. The ξ_i $(1 \leq i \leq n)$ are the coordinate functions on X: if $x \in X$, then $\xi_i(x)$ is the ith coordinate of x. $P(X)$ is generated as a k-algebra by the coordinate functions, and is called the coordinate ring (or affine algebra) of X.

As in Exercise 26, for each $x \in X$ let $\mathfrak{m}_x$ be the ideal of all $f \in P(X)$ such that $f(x) = 0$; it is a maximal ideal of $P(X)$. Hence, if $\tilde{X} = \mathrm{Max}(P(X))$, we have defined a mapping $\mu : X \to \tilde{X}$, namely $x \mapsto \mathfrak{m}_x$.

It is easy to show that μ is injective: if $x \neq y$, we must have $x_i \neq y_i$ for some i $(1 \leq i \leq n)$, and hence $\xi_i - x_i$ is in $\mathfrak{m}_x$ but not in $\mathfrak{m}_y$, so that $\mathfrak{m}_x \neq \mathfrak{m}_y$. What is less obvious (but still true) is that μ is surjective. This is one form of Hilbert's Nullstellensatz (see Chapter 7).

Solution. (I suppose the author wants me to prove μ is surjective.)

Recall that an ideal in the quotient ring is maximal iff the preimage of that ideal is maximal.

Denote the canonical map

$$\varphi : X \longrightarrow X/I(X)$$

So $\mathfrak{m} \in \tilde{X} = \mathrm{Max}\,(k[t_1, \ldots, t_n]/I(X))$ is a maximal ideal in $\tilde{X}$ iff $\varphi^{-1}(\mathfrak{m})$ is a maximal ideal in $k[t_1, \ldots, t_n]$ which contains $I(X)$.

Therefore, for any maximal ideal $\mathfrak{m} \in \tilde{X}$, since $\varphi^{-1}(\mathfrak{m})$ is a maximal ideal contianing $I(X)$, $\exists x \in X$, s.t. $\varphi^{-1}(\mathfrak{m})$ correspond to the maximal vanishing ideal of x. So $\mu(x) = \mathfrak{m}$.

Since $\mathfrak{m}$ is arbitarily chosen, we know μ is surjective.

Exercise 28

Let $f_1, \ldots, f_m$ be elements of $k[t_1, \ldots, t_n]$. They determine a polynomial mapping $\phi : k^n \to k^m$: if $x \in k^n$, the coordinates of $\phi(x)$ are $f_1(x), \ldots, f_m(x)$.

Let X, Y be affine algebraic varieties in k^n, k^m respectively. A mapping $\phi : X \to Y$ is said to be regular if ϕ is the restriction to X of a polynomial mapping from k^n to k^m.

If η is a polynomial function on Y, then $\eta \circ \phi$ is a polynomial function on X. Hence ϕ induces a k-algebra homomorphism $P(Y) \to P(X)$, namely $\eta \mapsto \eta \circ \phi$. Show that in this way we obtain a one-to-one correspondence between the regular mappings $X \to Y$ and the k-algebra homomorphisms $P(Y) \to P(X)$.

Solution. (See [Abstract Algebra (third edition) - Dummit & Foote])

$$\begin{array}{ccc}
X \subseteq \mathbb{A}_k^n & \xrightarrow{\quad\phi\quad} & Y \subseteq \mathbb{A}_k^m \\
\vdots & & \vdots \\
\mathbb{A}_k^n & \longleftarrow & \mathbb{A}_k^m \\
\downarrow /I(X) & & \downarrow /I(Y) \\
P(X) = k[t_1, \ldots, t_n]/I(X) & \xleftarrow{\quad\Phi\quad} & P(Y) = [t_1, \ldots, t_m]/I(Y)
\end{array}$$

First, for any regular mapping $\phi : X \to Y$, we have the induced k-algebra homomorphism $P(Y) \to P(X)$ by $\eta \longmapsto \eta \circ \phi$. We'll prove such correspondence is bijective, i.e. for all k-algebra homomorphism $\Phi : P(Y) \to P(X)$, there exist a regular mapping $\phi : X \to Y$ such that Φ was induced by ϕ.

For any k-algebra homomorphism ϕ, consider $\overline{t_1}, \ldots, \overline{t_m} \in P(Y)$ and their images

$$F_1 = \Phi(\overline{t_1}), \ldots, F_m = \Phi(\overline{t_m})$$

Now $(F_1, \ldots, F_m)$ defines a regular mapping $X \to Y$. Since the $P(Y)$ was generated by $\overline{t_1}, \ldots, \overline{t_m}$, any k-algebra homomorphism $P(Y) \to P(X)$ can be expressed by $\overline{t_1}, \ldots, \overline{t_m}$, in this way we obtained the required regualr mapping

$$\phi = (F_1, \ldots, F_m)$$